Precision-fermented dairy, egg and serum proteins: what they cost, and where evidence would help

Memo · v4.1, seven proteins (2026-10-06 11:00) · Pablo Antonio Moreno Casares · techno-economic model `pf-tea`, Humbird (2021) method, checked against NREL, Intelligen and Voutilainen designs · prepared with an AI assistant; not yet reviewed by human experts

Bottom line.

- **Today's public strains cost far too much.** At the best peer-reviewed titers, β-lactoglobulin (BLG, whey), ovalbumin (OVA), ovomucoid (both egg white) and β-casein cost $43.0, $227, $479 and $803 per kg of protein. The proteins they replace sold for $9–$18 (whey), $10–$21 (dried egg white) and $8–$19 (casein) per kg of protein in 2015–2025 (2025 dollars). Only one public titer comes near the design point, and a company reports it: EVERY's 30 g/L ovomucoid patent example, at $23.7.
- **At the titers in companies' own cost models, the commodity proteins near their markets.** At 42 g/L, BLG costs $15.8 ($12.3 with the lighter accounting of NREL-style design reports), OVA $15.9, ovomucoid $16.0 and β-casein $16.4. Whey sold above the BLG figure in 3 of 21 annual export prices in 2015–2025. Casein has the thinnest evidence: no secreted casein titer is public.
- **Secretion, not titer, is the gate.** To reach $20/kg, cells must secrete about 5–7 mg of protein per g of cells per hour at the base cell density. The base design secretes 4.7. With today's biology and capital, the best design reaches $7.1/kg BLG at 40 mg/g/h and $12.0 at 10. No published **K. phaffii** process comes near either rate; the best glucose fed-batch we found reached 2.1.
- **Sugar is not the problem.** Glucose makes up 19% of the base cost ($4.1/kg BLG). That is no more than an animal's feed per kg of protein, with all the feed charged to protein ($4.7–$6.6 in the US). The plant makes the gap: $17/kg, against $2–$8 on the farm.
- **A best case, not a floor.** Set every assumed limit at its best, with 400 m³ vessels and 100 kt/yr. If cells secrete 40 mg/g/h, BLG, OVA, ovomucoid and β-casein then cost $4.1, $4.2, $4.3 and $4.5 per kg. At 10 mg/g/h BLG costs $8.1 at the best rate shown so far (2.1 mg/g/h), $24.5.
- **Scarce proteins are the easier case.** Intelligen's lactoferrin design reaches 10 g/L with methanol. At that titer, a 250 t/yr plant makes lactoferrin for $130–$168/kg ($177 costed on methanol). That is below the $250–$1,585/kg reported for dairy lactoferrin. At the best public methanol-free titer (2.8 g/L) it costs $413, inside that range.
- **The route can change the product.** Precipitated, as dairy casein is, β-casein reaches dairy purity at $22.0/kg. The only public yeast casein process uses membranes alone. Its powder is about 6% β-casein and mostly yeast carbohydrate.
- **For cultivated-meat albumin, the columns set the cost.** We use the industrial process's minimal train (cation exchange, then anion exchange), cell-culture purity and a 1 kt/yr plant. Albumin then costs $215/kg ($205 at 42 g/L; $424 at the best public titer, 8.7 g/L of broth made with methanol). Bulk bovine albumin sells for $40–$70, and GFI targets $10/kg. Capture alone would cost $143, and a $100/L resin $114. Membranes alone cost $34.3 but reach only 77% albumin. At the base cost, albumin adds $1.95–$10.23 per kg of cultivated meat, and leading firms are removing it from their media.
- **For α-lactalbumin and κ-casein, the strains are missing.** α-Lac, the infant-formula whey protein, costs $33.8/kg at the base titer on a 1 kt/yr plant. Reports put α-lac at $25–$129 per kg (grade C). At its best public titer (0.6 g/L of supernatant, human α-lac) it costs $1,848. Rennet cheese needs κ-casein at about 19% of its casein. At the best κ titer (0.25 g/L), an α_{s1} + κ blend costs $567–$573/kg. To add under 10%, κ co-precipitated with α_{s1} must reach roughly α_{s1}′s design titer. Heme from the best secreting strain, used at 0.45%, adds $1.09 per kg of beef analogue at the factory gate.

$/kg protein, US	BLG	OVA	ovomucoid	β-casein	lactoferrin	albumin	α-lac
best public academic strain	$43.0	$227	$479	$803	$413	$424	$1,848
base case (30 g/L; lactoferrin 10 g/L)	$21.2	$21.3	$21.3	$22.0	$168	$215	$33.8
base case, design-report capital	$16.4	$16.5	$16.5	$17.0	$130	$185	$25.0
industry benchmark (42 g/L)	$15.8	$15.9	$16.0	$16.4	$106	$205	$28.2
physical-limit best case	$4.1	$4.2	$4.3	$4.5	$56.9	$172	$5.2
incumbent, 2025$ per kg protein	$8.7–$17.8	$9.7–$21.2	$9.7–$21.2	$7.9–$18.7	$250–$1,585	$40.0–$70.0	$25.0–$129

Plants: 10 kt/yr (lactoferrin 250 t/yr, albumin and α-lac 1 kt/yr); physical limit at 100 kt/yr (lactoferrin 1.6 kt/yr, α-lac about 19 kt/yr). Albumin's incumbent: bulk bovine albumin (technical grade).

Where evidence would help most, in our judgment:

1. **Secretion without growth.** How much protein does a strain secrete per g of cells per hour when it barely grows, over days, at 1–10 m³? This rate sets both today's cost and the best case, for every protein here.
2. **Open fed-batch data.** Titer, productivity and sugar yield, measured **together**. For casein: any secreted titer above a few g/L, and whether the unphosphorylated protein makes cheese like dairy casein. For bovine α-lac and κ-casein: any secreted titer above 1 g/L.
3. **Real plant costs.** The capital, staff and running costs of recent PF plants would settle the vessel price and the fixed-cost rule. Aseptic vessels would add $9.8/kg, and the two accounting conventions differ by $4.7/kg on BLG.
4. **The design ceiling.** Workable starting volumes, feed concentrations and O_2 transfer at scale.

Animal welfare. The cost model does not say which protein matters most for animals, but the products differ. Whey is a by-product of cheese, so replacing it mostly lowers the whey price. Casein is what people buy milk for. Egg white (OVA, ovomucoid) carries far more animal-years per kg of protein than either.

EU. At Dutch prices for sugar, energy, wages and construction, the BLG plant costs about $3.1/kg more. Capital at 4% over 20 years, instead of 7.5% over 10, would cut $3.9/kg. These are cost effects on one plant, not policy recommendations.

Caveats. These are production costs with a 14.6%/yr capital charge, not market prices. They exclude R&D, licensing, regulation, sales, start-up, failed batches and the premium of a first plant. Strain parameters rest on design values and company-reported averages (evidence grades B–C). The casein and lactoferrin titers come from other proteins or species. Only AI reviewers have audited the work since version 2. Full report, audit log and data: the `pf-tea` repository.